

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1206

West Dallas Tap 138 kV
March 24, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1206. The Interconnection Customer proposes a 93.6 MW solar photovoltaic facility interconnecting at the West Dallas Tap 138 kV substation in the Pennsylvania area. The request was received on May 17, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 483 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$37,295,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Alta – Dearborn Industrial Generation 138.0 kV: 127.8% loading (198 MVA) following the outage of the Alta – Lakeshore Mine 138.0 kV; DFAX = 64.1%.

For the loss of Alta – Dearborn Industrial Generation 138.0 kV L, the Lakeshore Mine – Alta 138.0 kV loads to 117.4% of its 201 MVA rating. Project DFAX: 59.6%.

Hills Chapel 161-138 kV Auto: 110.1% loading (236 MVA) following the outage of the Alta – West Dallas Tap 138.0 kV; DFAX = 70.0%.

For the loss of Burns – Hills Chapel 345.0 kV, the Alta – W Dallas Tap 138.0 kV loads to 106.5% of its 374 MVA rating. Project DFAX: 67.2%.

Lakeshore Mine – Graham 138.0 kV: 104.6% loading (206 MVA) following the outage of the Alta – Dearborn Industrial Generation 138.0 kV L; DFAX = 59.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alta – Dearborn Industrial Generation 138.0 kV	Rebuild line with higher-capacity conductor	\$10,137,000	91% (\$9,193,000)
Alta – Lakeshore Mine 138.0 kV line	Rebuild line with higher-capacity conductor	\$15,912,000	91% (\$14,430,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hills Chapel 161-138 kV Auto	Replace transformer with higher-rated unit	\$8,674,000	83% (\$7,169,000)
Alta – W Dallas Tap 138.0 kV	Reconductor line	\$5,996,000	90% (\$5,415,000)
Lakeshore Mine – Graham 138.0 kV	Reconductor line	\$1,200,000	91% (\$1,088,000)
	Total allocated to project		\$37,295,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.